Dendroarchaeology

Lesson 5 · 90 minutes · From wood dates to human history

Learning objectives

Explain the questions addressed by dendroarchaeology.

Distinguish outer-ring, felling and construction dates.

Use preservation and context to state conclusions and limits.

From measurement to archaeological interpretation

Crossdating assigns calendar years to preserved rings.

The last measured ring may not be the original outermost ring.

Now interpret the dates in their archaeological context.

Ring years and event dates

01 **Ring formation**

02 Felling

03 Working / storage

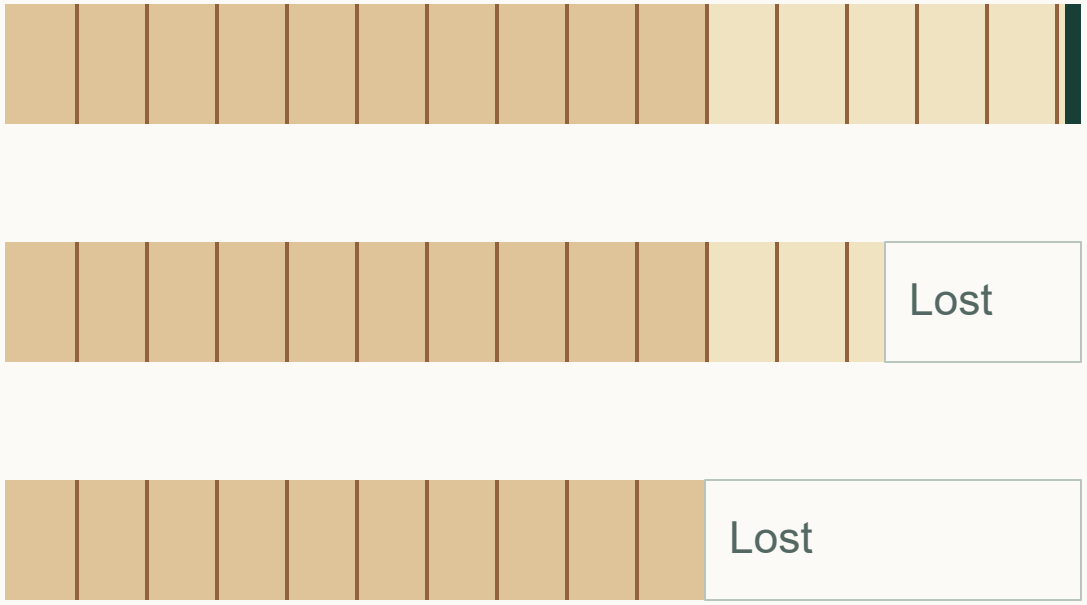
04 Construction / use

05 Repair / reuse / deposition

Conceptual sequence; intervals are not drawn to scale.

Inspect surviving outer wood

Pith side → bark side



Bark edge survives: approach the final growth increment.

Some sapwood survives: missing outer rings may be estimated.

Heartwood only: a precise felling date is harder to establish.

Original schematic / fictional teaching data, not empirical measurements.

What does a surviving bark edge establish?

Inspect the outermost increment beneath the bark.

Check incomplete growth and possible cutting after tree death.

Relate final growth to felling only with supporting conditions.

Estimating missing sapwood

Identify the heartwood–sapwood boundary.

Use a sapwood model appropriate to species and population.

Do not transfer another region’s ring-count range uncritically.

Interpreting a lower date bound

A terminus post quem is a lower bound, not an upper bound.

A last surviving ring of 1580 excludes felling before 1580.

Unknown missing rings leave the upper bound unresolved.

Considering the felling season

Examine earlywood and latewood development.

Growth timing varies with species, site and climate.

A winter interval may span two calendar years.

Sampling within a building

Link member position, joints, tool marks and repair records.

Seek preserved outer wood and multiple members.

Consider whether different members came from the same tree.

Exercise A: how tightly can you date it?

ID	Preservation	Dates / assumptions
A	Bark edge; incomplete outer ring	1600; assume cut while alive
B	Heartwood; outer loss unknown	Last retained ring: 1580
C	5 sapwood rings; assume 15–25 total	Last heartwood ring: 1590

Task: state felling and construction inferences, assumptions and limits.

Original schematic / fictional teaching data, not empirical measurements.

Exercise A: intended interpretations

A: felling in the 1600 growth season, under the stated conditions.

B: felling in or after 1580; no upper bound established.

C: approximately 1605–1615, conditional on the supplied model.

Members are not necessarily independent trees

Several boards or beams can come from one tree.

Compare patterns, anatomy and member information.

Distinguish possible same-tree matches from independent dating evidence.

Compare date groups with building phases

Plot dates together with member locations.

Compare them with proposed building phases.

Ask whether younger timber marks repair, extension or replacement.

Interpreting older timber

Reused timber can predate its current structural use.

Inspect redundant joints, tool marks, dimensions and location.

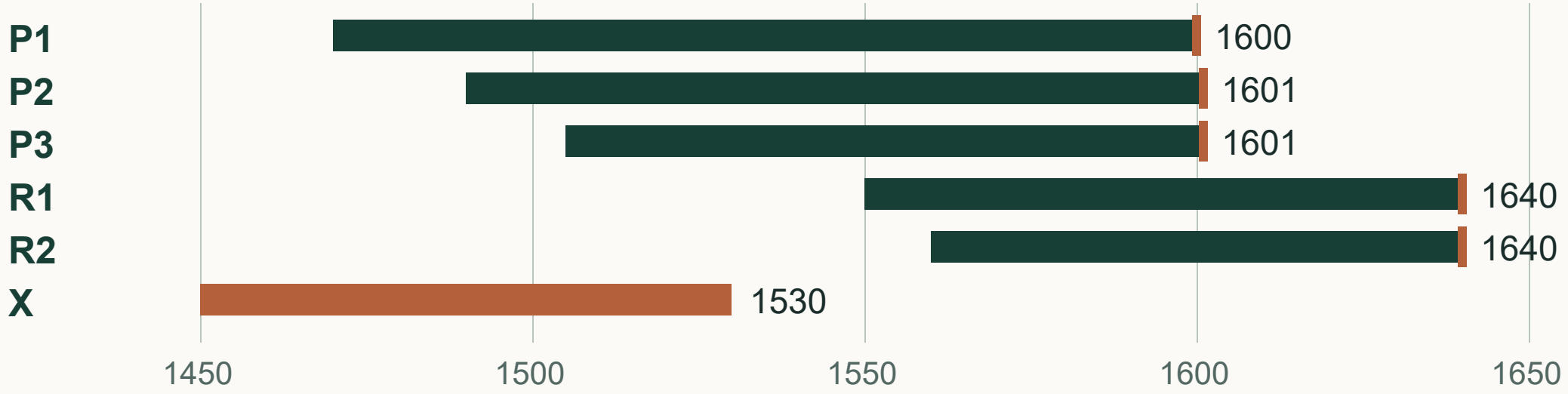
An older date alone does not prove reuse.

Exercise B: construction, repair or reuse?

Main frame P1–P3: complete outer edges, ending 1600, 1601 and 1601.

Roof R1–R2: complete outer edges, ending 1640.

Beam X: heartwood ending 1530 and a redundant joint.



Original schematic / fictional teaching data, not empirical measurements.

Exercise B: conclusions and alternatives

The sampled main-frame arrangement cannot be completed before 1601.

The 1640 roof timbers suggest a later intervention to investigate.

X is a reuse candidate; missing outer wood and other explanations remain.

Timber provenance is a separate question

Dendroprovenancing compares candidate source regions.

The best match need not identify a unique origin.

Consider reference coverage, timber trade, species and anatomy.

Reading a Japanese report: Horyuji

Mitsutani (2001) reports a 594 CE felling date for the central pillar.

Three further outer rings extend the measured sequence ending in 591.

What evidence connects a member date to the date of the building?

Yakushi-ji East Pagoda: dates by timber

The surviving outer surface affects what a date means.

Timber	Surviving outer wood	Outermost ring
Central pillar, lower part	Sapwood; no bark	719 CE
First-storey backing boards (2)	Bark present	729 / 730 CE

The bark-edge boards indicate felling dates; the pillar lacks its outermost wood.

Combine ring dates with historical and architectural evidence.

Hoshino, Kojima & Mitsutani (2017), pp. 75–77

Reading an English report: Rook Hall Farm

Complete sapwood in one sample supports winter 1522/23 felling.

The report suggests construction in 1523 or within a year or two.

Separate the felling result from the construction interpretation.

Record undated material as well

Short series, weak patterns or missing references can prevent dating.

Dated members may not represent the whole structure.

State what remains undated and propose further observations.

Evidence to include in a report

Specimen ID, location, species and outer-edge preservation.

Reference, overlap, candidate dates, mismatches and checks.

Type of conclusion, assumptions, alternatives and references.

Exit questions

Does heartwood ending in 1580 establish construction in 1580?

Where do you add a total sapwood estimate if five rings already survive?

Give a reason and one useful additional observation for each.

Next: reading forest history

Lesson 6 introduces dendroecology.

Study suppression, release, disturbance and regeneration.

Keep distinguishing tree dates from event dates.